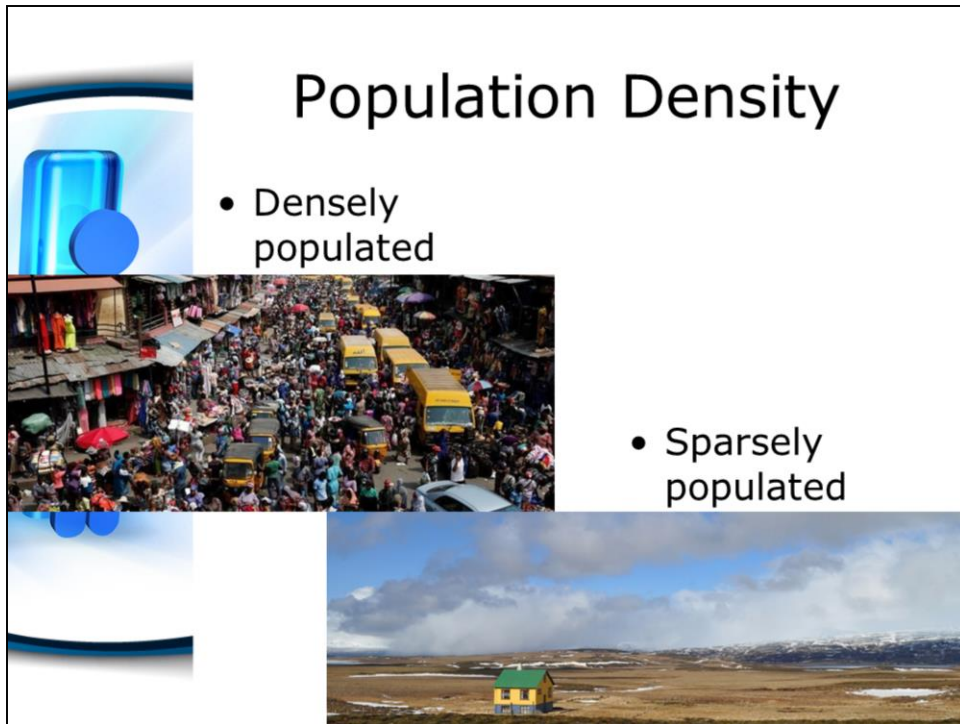


Population Distribution

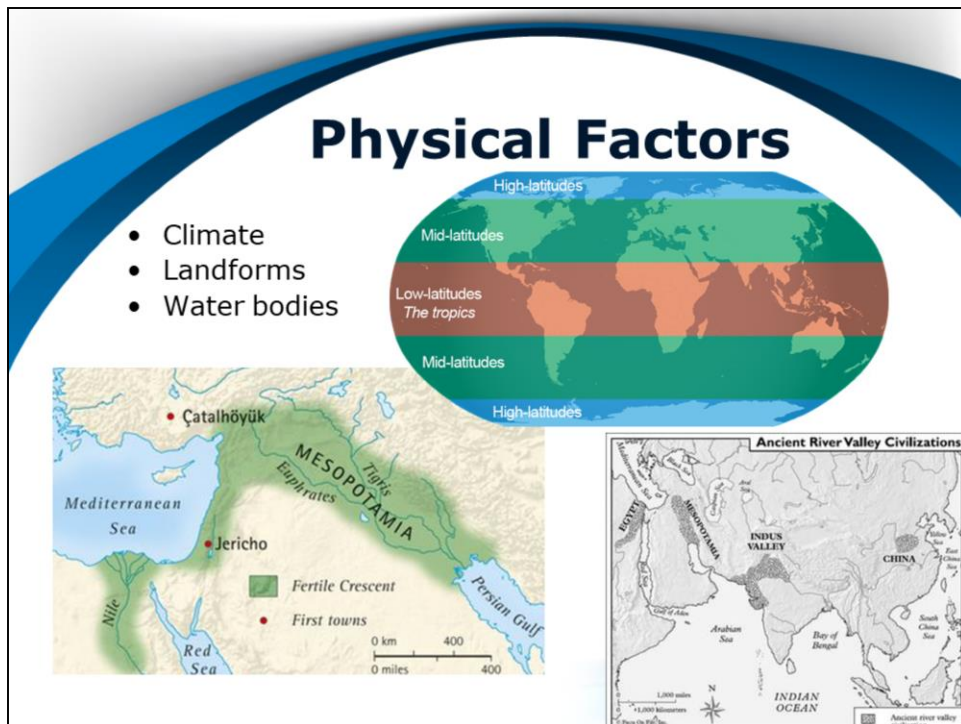


RRHS APHG 2019-2020



Is a place crowded? Or does it seem empty? Urban areas (cities) are often very densely populated. People seem to be living on top of one another, or they literally are if they live in high-rise apartment buildings. If you travel through a sparsely populated area, it may feel like you go miles and miles without seeing another person.

What causes populations to cluster where they do?



To understand population distribution patterns, we must first look at how major human settlements came to be. For the vast majority of human existence, humans were nomadic and tribal. About 12,000 years ago, at the time of the Neolithic Revolution, tribes quit following the food (hunting and gathering) and instead figured out that they could grow crops and raise animals as food deliberately– and thus began the practice of “agriculture”. With agriculture came permanent settlements in the places with the physical geographic site features most favorable for their new economic activity – farming! These early settlements formed in the world’s great river valleys – the land was relatively flat (plains), the climate mild (not too hot, not too cold, and moderate amounts of rain fall during the growing season), and near sources of freshwater that provided both fertile soil and transportation routes.

Common Physical characteristics of population clusters:

Climate – Early population centers developed in the **mid-latitudes** – These climate regions are shown in Green on the world map above. These are the zones that are not too hot (like the tropics) nor are they too cold (like the high-latitudes/polar zones). The mid-latitudes, also known as the “temperate” regions have distinctive summers and winters, and overall much more moderate climates than either the tropics or the polar areas. Rainfall must be moderate, as well – the tropics receive too much rain and therefore the soil is

leached of its nutrients, making it unsuitable for farming for large populations. Arid deserts (both hot such as the Sahara and cold such as the Gobi Desert or Siberia) do not receive enough rainfall to sustain crops. Regions too far north/south of the equator do not have long enough growing seasons (the time between the last frost of spring and the first frost of autumn) for plants to develop and produce food before the plants would freeze and die, and the land in polar regions is frozen virtually all year.

Landforms – Populations were dependent on farming for survival, so they lived on land that was easily farmed – which means it needed to be relatively flat. River valleys and the low-lying coastal plains, especially where rivers meander across the land carrying rich, organic sediment that was spread over the land during annual flood events, were the ideal for the development of these early population centers.

Water bodies – Civilizations tended to grow up in river valleys for a number of reasons. The most obvious reason is access to a usually reliable source of water for agriculture and human needs. Plentiful water, and the fertilization of the soil due to annual floods, made it possible to grow surplus crops – beyond that needed to sustain small agricultural villages. This surplus made the development of cities possible (more about this topic on the next slide, “human factors”).

The “zoomed in” map on the left above shows Mesopotamia – thought to be the earliest large population center and the **hearth** (point of origin/creation) of most European culture, from food to languages to religion.

The black-and-white map above shows the locations of the earliest civilizations, which were the first to be large population centers and the places from which culture developed. From West to East they include: The Nile River Valley, Mesopotamia (which combined with the Nile create the area known as the “Fertile Crescent” of Southwest Asia), the Indus River Valley (modern-day India and Pakistan), and the Yellow River Valley (modern day China).



Human Factors

- Culture
- History
- Economics
- Politics



The fertile river valley soil enabled farmers to grow a surplus of food, saving enough to get them through not only the winter, but also times of prolonged drought. This surplus also allowed for some members of the community to engage in non-agricultural activities such as construction of buildings and infrastructure, metal working, trade, and social organization. Boats on the river provided an easy and efficient way to transport people and goods, allowing for the development of trade and facilitating central control of outlying areas. The emerging urban centers had a high degree of “centrality” – they served as a gathering point for economic, cultural, and political activities for the surrounding areas.

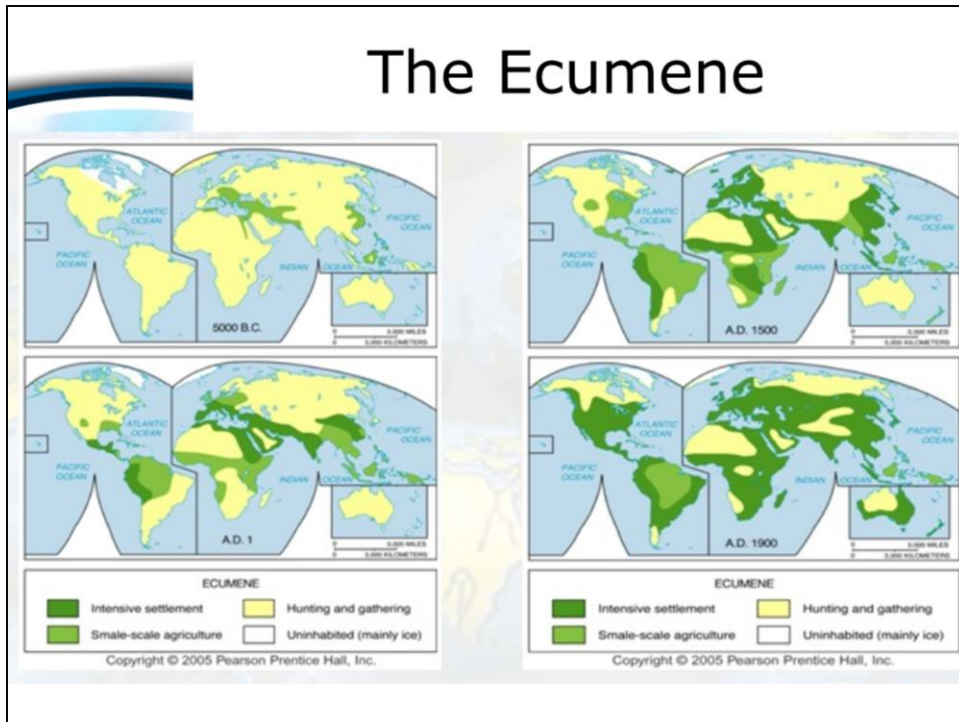
Culture – urban centers serve as culture centers – places of religion, arts, education, etc. attract people to them and influence the areas around them. Culture also influences the number of children born into families, and as cultures shift from traditional to more modern, the number of children per family tends to decrease. The status of women within society also influences the **total fertility rate** (the average number of children born to each woman during her lifetime), as increased access to education for a woman generally leads her to have fewer children.

Economics – urban centers serve as hubs of economic activity. The more powerful the economic influence of the city, greater the pull as a major population center as people go there to find jobs in manufacturing and services. These urban centers will be much more densely packed with people than the surrounding agricultural areas. Again, the status of women in the workforce also effects total fertility rates, as women who pursue a career outside the home tend to delay the start of their family and have fewer children.

History – the early population centers had large numbers of people early on, and continue to be among the most populated places today. Cities of historical significance (such as London, Paris, Rome, etc.) also maintain large populations today.

Politics – politics influence population distribution. Historic cities were dependent on protection of their feudal lord and his army of knights. Urban areas tend to benefit most from the provision of government services, such as schools, parks, public transportation, hospitals, etc. Political leaders can also influence how many children women have by either encouraging women to have children by providing incentives such as extended paid maternity leave and tax breaks per child; or, conversely, they can discourage women from having more children by funding birth control programs or even passing laws putting limits on the number of children families can have (such as the “One Child” policy that was enforced in China’s urban areas by fining families that had more than one child and restricting subsequent childrens’ access to government services such as school and healthcare).

The Ecumene



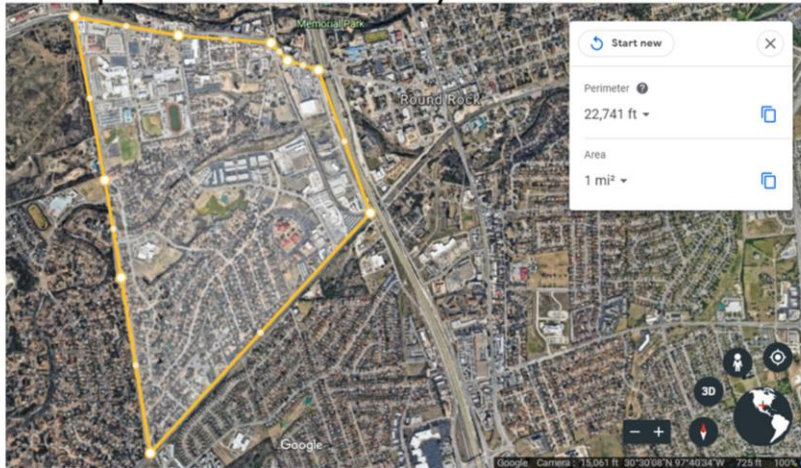
Ecumene was an ancient Greek term for the known, the inhabited, or the habitable world. The term is used by geographers today to mean permanently inhabited land. It generally refers to land where people have made their permanent home, and to all work areas that are considered occupied and used for agricultural or any other economic purpose.

The maps above show the expansion of the Ecumene in 5,000 B.C., A.D. 1, A.D. 1500, and A.D. 1900. The dark green areas represent “intensive” land settlements – meaning large populations who put a significant strain on the environment to meet human needs. Notice in A.D. 1, the dark green areas comprised of the world’s major river valley settlements, which had already been in existence for about 10,000 years at that point, plus the Aztec, Incan, and Mayan civilizations in the Americas. Meanwhile, small-scale (subsistence and less-intense farming of smaller populations) was expanding into Southeast Asia, the Sahel region of Africa (the fertile area between the Sahara Desert and the African rain forest), and the plains of Northern South America, and the Gulf Coastal Plain of what is now the United States. By A.D. 1500, hunting and gathering regions of the world (not considered part of the Ecumene, because hunting and gathering is NOT permanent, nor do the nomadic tribes alter the land to intentionally grow food because they do not practice agriculture). The ecumene of A.D. 1900 consists of pretty much everything except the world’s

largest deserts or tundras. Hunter and gather tribes only exist in very small isolated areas by this point.

Population Density

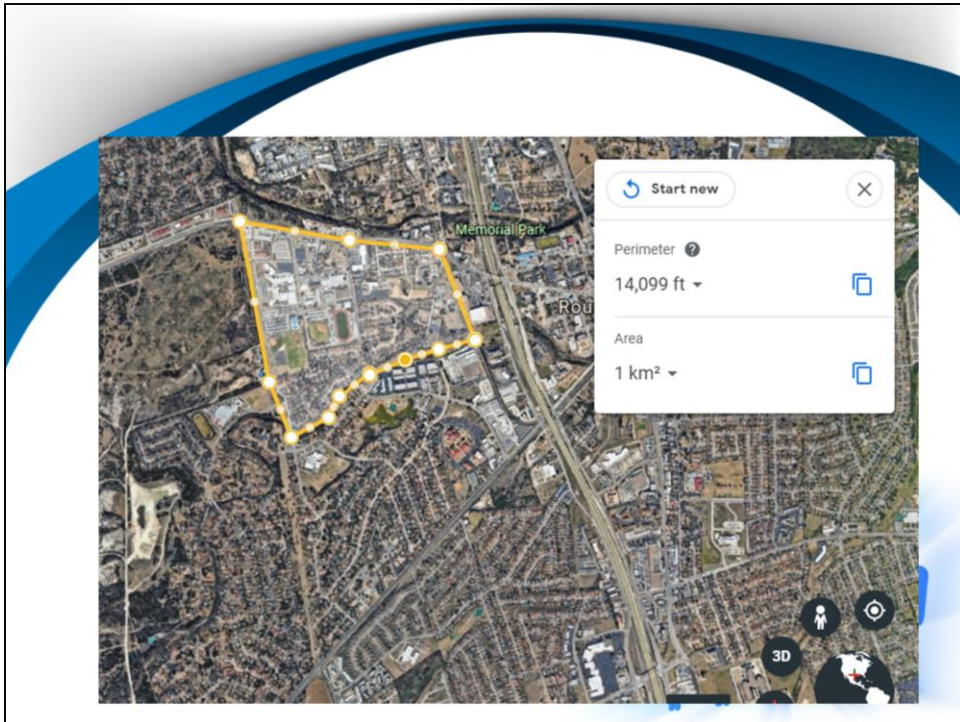
- Population divided by units of land



Population Density is a calculation of how many people live within a certain unit of land. The value for unit of land will be different depending on the characteristics of the land we are considering (i.e., total land area, vs. units of arable land).

For a frame of reference, I have included a map of Round Rock with 1 square mile highlighted. It starts at the upper left corner at the intersection of Deepwood Drive and R.R. 620, over to I-35, south to McNeil Road, over to the Austin Stone campus, and then connects back to 620 via the Round Rock West Green Belt/Deepwood Drive. When we are talking about population density per unit of land in square miles, this is the amount of land to which we are referring. It is an area about 5x the size of the Round Rock HS campus.

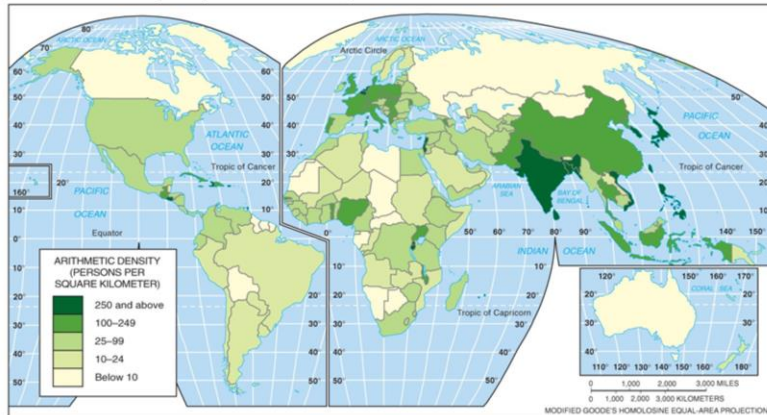
You may sometimes see the unit of land in square kilometers – which is just slightly more than 1/3 of a square mile's area (see next slide for comparison of area).



For comparison, the polygon on this screen shot shows a square kilometer – It is an area about 2x the size of the Round Rock HS campus.

Arithmetic Population Density

- Total population/total land area



Arithmetic population density is the simplest of density statistics. To calculate it, all you need to know is the number of people that live in the area and the total amount of land that comprises the area.

Arithmetic population density = total population/total land area

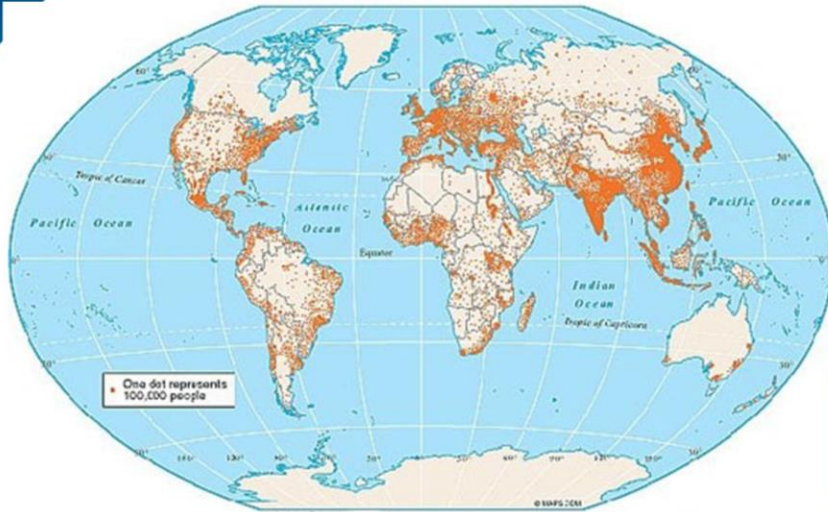
(note the map above map is displaying arithmetic density in population per square kilometer (the rest of the world uses Metric units of measurements, so most maps will display the data by square kilometers)

Also, pay attention to the population density of the areas where the world's earliest population centers developed – remember, they had a head start on building population there, so they area still densely populated today.

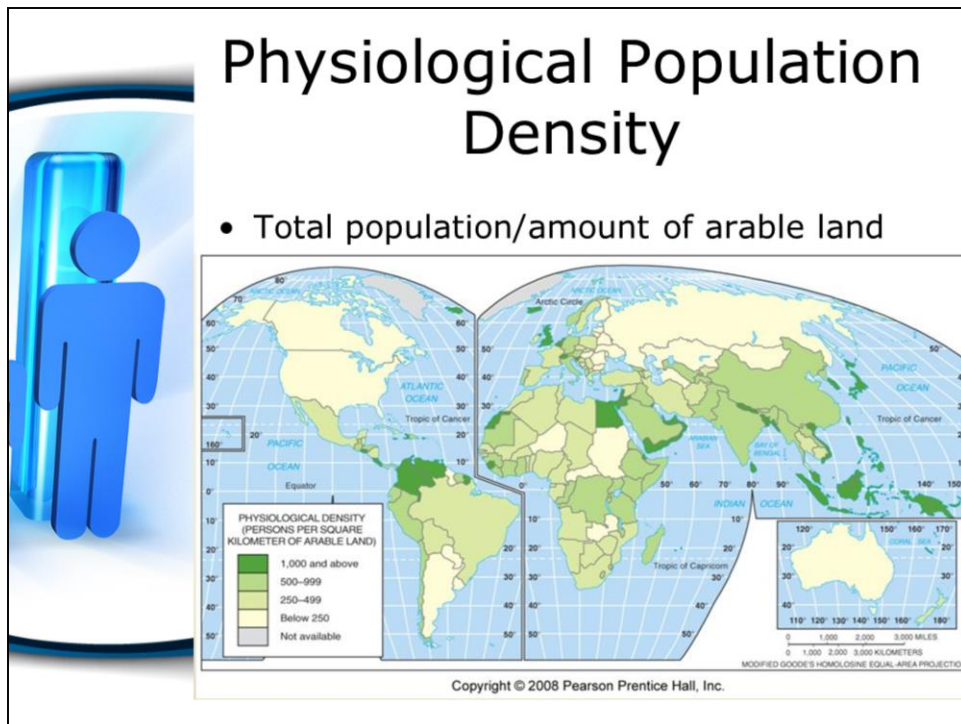
How does the scale of analysis (the geographic unit for displaying data is “country”) affect your interpretation of the information? If the scale of analysis changed to either square miles OR the geographic unit changed to show internal units such as provinces/states, would you see equal distribution within the same countries? What about if the map were dot density rather than

choropleth? (see next slide for an example)

Population Density

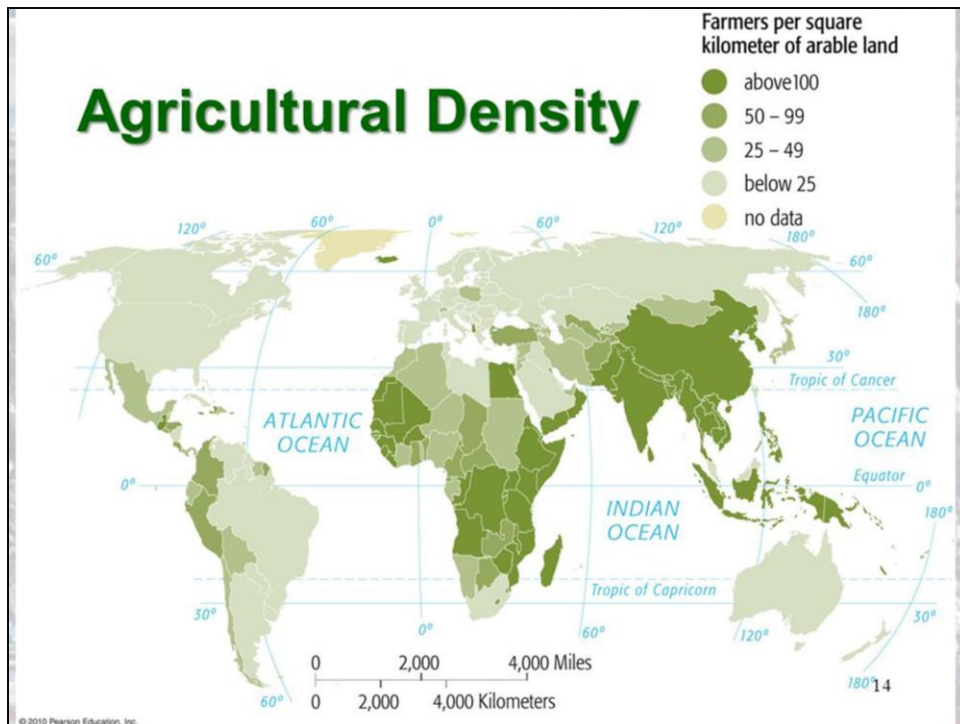


The scale of analysis on this dot-density map is 1 dot = 100,000 people. With this map, we are able to see the unequal distribution of people within the countries.



Physiological population density is another measure of population density that calculates the total population divided by the amount of **arable** land. This statistic is used to analyze the amount of stress put on the land to meet the needs of the population. As such, geographers often use this measurement to estimate how long the land and its resources can support its population (this is known as **population pressure**).

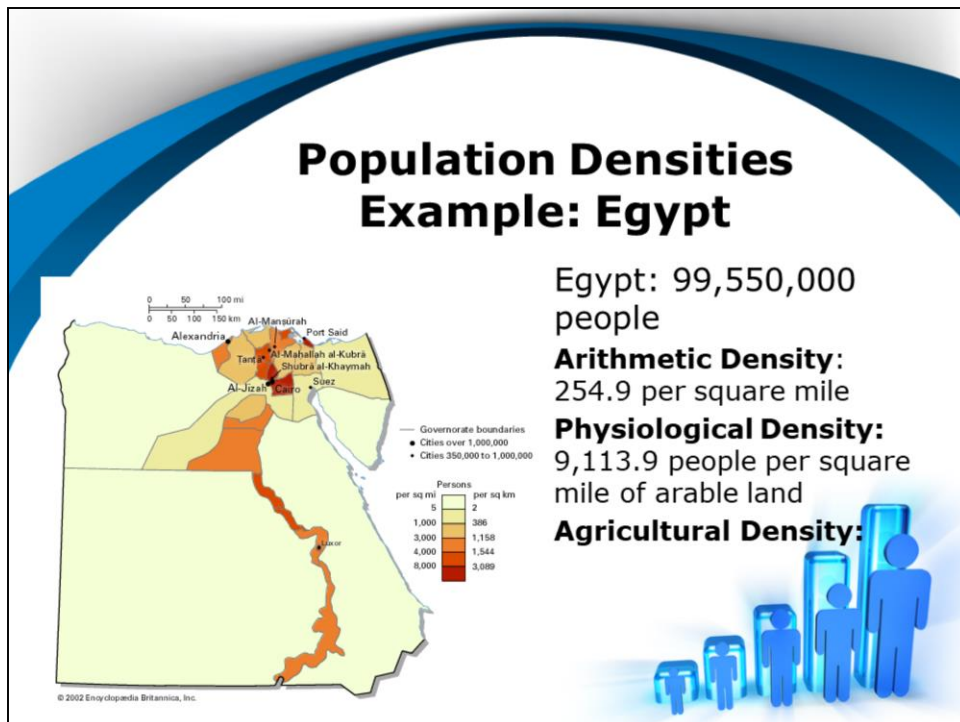
If you compare this map to the map from 2 slides ago (arithmetic density), you can see that Egypt's physiological population density is much higher than that of the United States, while their arithmetic density is similar. This means that Egypt's population likely puts much more pressure on the land and will likely experience resource issues related to arable land sooner than countries with a lower physiological density.



Agricultural density is a measure of the number of farmers per unit of arable land.

Agricultural density = #of farmers/units of arable land.

A high agricultural density suggests that available arable land is being used by more farmers and may reach the limit of its production potential sooner than where agricultural density is lower. It is another measure of population pressure.



Total population of Egypt (2017): 99,550,000.

Total land area of Egypt: 390,100 mi²

Calculating the Arithmetic population density of Egypt.

To figure out this statistic, we must know the population of Egypt and the amount of land that comprises the country in square miles, the two counts provided above. So, the arithmetic population density of Egypt is = $99,550,000 / 390,100 \text{ square miles} = 254.9 \text{ people per square mile}$ (this is a low density).

Calculating the Physiological population density of Egypt:

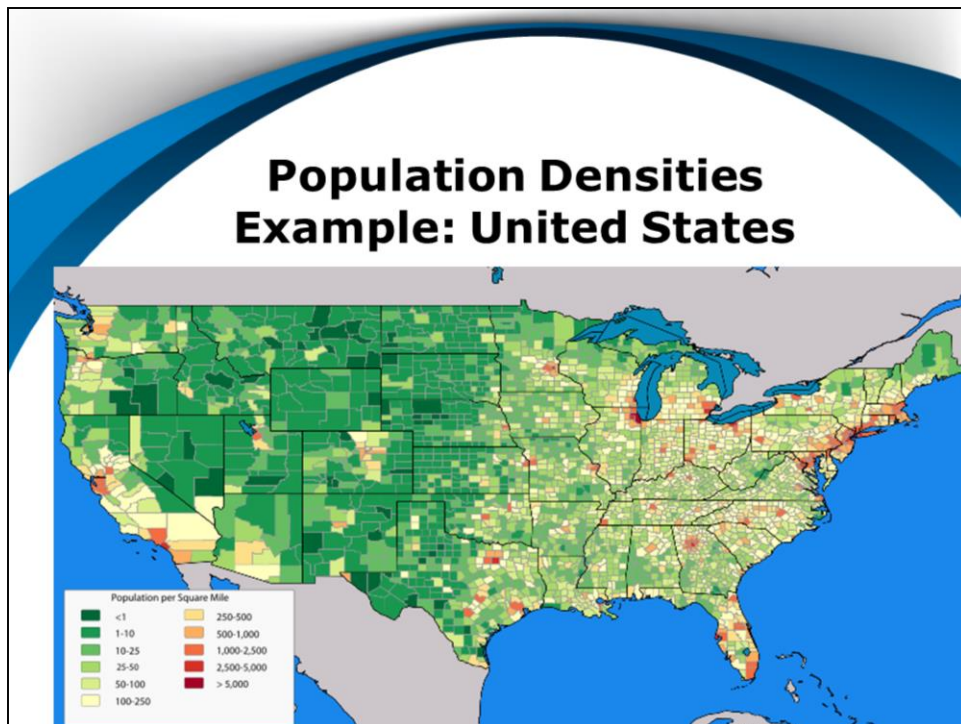
To calculate this statistic, we need to know the amount of arable land in Egypt. According to the CIA World Factbook, only 2.8% of Egypt's land is arable. If we multiply 390,100 sq. mi. by 2.8% (.028 on your calculator), that means Egypt has 10,922.8 sq. miles of arable land. That's not very much land to feed almost a hundred million people.

$99,550,000 / 10,922.8 = 9,113.9 \text{ people per square mile of arable land}$. This is a very high physiological population density.

Calculating the Agricultural population density of Egypt:

Now we need to know both the number of farmers in Egypt AND the amount of arable land. According to the CIA world Factbook, 25% of Egypt's labor force works in agriculture, so we multiply 99,550,000 by 25% (.25 on your calculator) to determine there are 24,887,500 farmers in Egypt (that's only slightly less than the population of Texas, by the way).

Agricultural density = 24,887,500 farmers / 10,922.8 sq. miles of arable land = 2,278.5 farmers per square mile. Considering there are 640 acres in a square mile, that means each farmer gets about 3.5 acres to farm (if it were evenly distributed, of course).



Total population of United States: 329,257,000 (2018 estimate)

Total land area (sq mi) of the United States: 3,797,000 mi² (rounded)

Total arable land area (sq. mi) of the United States: 16.8% (3,797,000 x .168)
= 637,896 sq. mi. of arable land

of farmers in the United States = 0.7% of the population x 329,257,000 =
2,304,799 farmers in the United States

Agricultural Density = $329,257,000 / 3,797,000 = 86.7$ people per sq. mi. of land

Physiological Density = $329,257,000 / 637,896 = 516.5$ people per sq. mi. of arable land

Agricultural Density = $2,304,799 / 637,896 = 3.6$ farmers per sq. mi. of arable land (or about 178 acres per farmer)

The United States has a low population density based on all three measurements. This means that the population pressure on the land in the United States is much lower than that of Egypt.



Population Pressure is the sum of the factors (due to growing population, increase in consumption of resources, or environmental disaster) within a population that reduce the ability of an environment to support the population and that therefore tend to result in migration and expansion of range or in extinction or decline of the population.

Carrying Capacity (in terms of human geography) is the number of people that a region can support without environmental degradation. If the population pressure exceeds the carrying capacity, the population will either need to expand their range (which may cause conflict with surrounding countries or push disadvantaged populations off their land) they will need to migrate to a new location.

Over-population occurs when a population's size or consumption of resources exceeds the carrying capacity of the land and results in damage to the land, a lack of resources, or even conflict (war) as people fight over control of the limited resources, such as water, farmland, or non-renewable resources such as oil/natural gas. A very common cause of overpopulation in cities is poverty as there are not enough services (human resources) to adequately meet the needs of the poor in many places around the world. Overpopulation

can also occur if agricultural land is used for purposes other than food (agricultural land is a resource, and this would be a reduction of a valuable resource). This situation is happening in many parts of the world today as farmers in less developed nations shift to more profitable crops such as cash crops (i.e., tobacco, cotton, or export luxury crops such as cocoa or coffee), leaving less land available for those farming food for the population. Additionally, as commercial farmers growing cash crops often seek to expand their growing operations, so they acquire more land, which pushes small-scale farmers (family farmers) off their land and either onto less desirable land (such as a rainforest, which has low soil quality due to leeching of nutrients in the soil) or into the rapidly growing cities. As these former farmers arrive in the cities, there are often not enough jobs for everyone, which contributes to the formation and persistence of slums. The farmers pushed into the rainforest must clear the land to farm it, which leads to deforestation, soil loss, and habitat loss.

Under-population occurs when a population's size is lower than normal or than is desirable. This means that there are not enough people to do the work necessary to sustain the population.

The unequal distribution of people within a country can lead to over-population of some areas and under-population of others. If such an imbalance occurs, the government may try to encourage people to move from the over-populated area to the under-populated area for political or economic reasons. For example, the percentage of farmers in the United States is very low. While machinery often works to make up for under-population in agricultural regions, some crops do not lend themselves to being harvested by machines and there simply are not enough people available in the area to harvest the crops. One approach to correcting this issue is to encourage migrant farmers or guest workers (immigrants with short-term permissions to work in the country) to work on the farms in that area during the harvest season. Underpopulation also makes the provision of services difficult. There is an economy of scale in densely populated areas that makes the cost per person much lower. For example, in education, schools in rural areas often do not have the variety of classes or access to technology that schools in urban areas have. This is because the cost per student to bring in specialized teachers for advanced classes, sports, or specialty subjects (such as ALS, AP classes, or even a football coach) would be too high. If there are only 50 students in an entire high school, they may only have 3 or 4 teachers total, compared with a high school with over 3,000 students that has 300 faculty members, which allows

for electives, advanced classes, and creation of degree plans (academies) that help students prepare for future careers. Over-crowded schools (common in poor areas of major urban areas) may not have enough resources to go around, which may result in a shortage of teachers who do not want classes with 50 students each, a lack of supplies